Machine Learning-Based QSAR Modeling for Prediction of PD-L1 Inhibitory Potency

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Abstract—**The PD-1/PD-L1 immune checkpoint pathway plays a critical role in cancer immune evasion. While antibody-based therapies exist, they are limited by cost, toxicity, and lack of oral availability. This study proposes a machine learning-based QSAR framework to predict PD-L1 inhibitory activity using molecular descriptors. Two datasets—ChEMBL (real-world) and GitHub PD-L1 (curated)—were used to train Random Forest, Gradient Boosting, and XGBoost models. Performance was evaluated using R^2, RMSE, and MAE. Results show that XGBoost achieved the best performance, particularly on the curated dataset ($R^2 \approx 0.65$), while performance on the ChEMBL dataset remained moderate ($R^2 \approx 0.45$). The study highlights the importance of dataset quality and descriptor selection in QSAR modeling.**

Index Terms—QSAR Modeling, PD-L1 Inhibitors, Machine Learning, Drug Discovery, Molecular Descriptors, XGBoost, Random Forest

I. INTRODUCTION

Cancer cells often evade immune detection by exploiting the PD-1/PD-L1 pathway, where Programmed Death-Ligand 1 (PD-L1) on tumor cells binds to the PD-1 receptor on T-cells, suppressing their cytotoxic activity. While monoclonal antibodies targeting this pathway have revolutionized cancer therapy, they suffer from significant drawbacks, including high manufacturing costs, limited tumor tissue penetration, and severe immune-related adverse effects. This has motivated a shift towards developing small-molecule inhibitors, which offer advantages such as oral bioavailability and better safety profiles. Discovering these inhibitors can be accelerated using computational methods. Quantitative Structure-Activity Relationship (QSAR) modeling, a powerful tool in drug discovery, enables the prediction of a molecule's biological activity from its structural and chemical features. This study develops and evaluates a machine learning-based QSAR framework for predicting PD-L1 inhibitory activity using molecular descriptors derived from two datasets with different levels of curation and variability. The predictive performance of Random Forest, Gradient Boosting, and XGBoost is

compared using R^2, RMSE, and MAE in order to assess the influence of data quality on QSAR model reliability.

II. RELATED WORK

Recent studies have demonstrated that computational approaches can significantly accelerate the discovery of small-molecule inhibitors targeting the PD-1/PD-L1 immune checkpoint pathway. In particular, machine learning, molecular docking, and molecular dynamics simulations have emerged as effective complementary strategies for

identifying compounds with favorable inhibitory potential while reducing the time and cost associated with purely experimental screening.

et al. investigated FDA-approved small molecules as potential PD-1/PD-L1 inhibitors by integrating docking-based virtual screening, molecular dynamics simulations, and experimental validation. Their study identified six candidate compounds with binding modes comparable to a reference ligand, and molecular dynamics analysis confirmed stable interactions with the PD-L1 protein. Among the shortlisted compounds, montelukast showed the most promising inhibitory behavior, demonstrating concentration-dependent activity in an ELISA-based assay. This work is important because it highlights the value of combining computational screening with experimental testing to validate potential PD-L1 inhibitors.

Tong et al. developed a machine learning-based virtual screening framework for the discovery of novel PD-L1 inhibitors using a dataset of 2044 experimentally validated compounds. After extracting molecular descriptors from RDKit and Mordred, they applied feature screening and trained multiple gradient boosting models, including LightGBM, GBDT, XGBoost, and CatBoost. Among these models, CatBoost achieved the best generalization performance, with an R^2 of 0.83 on the test set, and was selected for large-scale screening of lead-like compounds from the ZINC database. Their study further demonstrated that accurate activity prediction alone is not sufficient for identifying viable drug candidates. To improve practical relevance, they incorporated ADMET-based evaluation, including human intestinal absorption, MDCK permeability,

plasma protein binding, CYP3A4 and CYP2D6 interactions, hepatic and renal clearance, and toxicity-related assessment. This additional screening step narrowed the candidate pool to compounds with more favorable pharmacokinetic behavior and lower predicted toxicity, thereby showing that QSAR-guided discovery should be supported by drug- likeness and safety considerations.

Kuttappan et al. proposed a machine learning-assisted sequential virtual screening strategy to identify PD-L1 inhibitors from a large chemical library. Their workflow combined high-throughput molecular docking, ADME screening, molecular dynamics simulations, and Random Forest-based QSAR modeling using both Dragon descriptors and molecular fingerprints. The resulting models showed strong predictive performance and were able to estimate the inhibitory activity of newly screened leads. In addition, their analysis identified important structural features associated with PD-L1 inhibition, providing useful insight into the

QSAR modeling can improve both lead identification and mechanistic interpretation.

Taken together, these studies show that machine learning has become a valuable tool for QSAR-based prediction of PD-L1 inhibitory activity. They also indicate that model performance depends not only on the choice of algorithm, but also on data quality, descriptor selection, validation strategy, and downstream pharmacokinetic assessment. Therefore, the present study builds on this prior work by applying machine learning-based QSAR modeling to PD-L1 inhibitors and comparing model performance across two datasets with different levels of variability and curation quality.

III. METHODOLOGY

1) *Dataset Description*

Two datasets were used in this study to develop and evaluate QSAR models for predicting PD-L1 inhibitory activity. The first dataset was obtained from ChEMBL and consisted of experimentally reported bioactivity records with relatively high variability. Records were filtered for *Homo sapiens*, single-protein targets, and IC50 measurements associated with PD-L1 inhibition. After preprocessing and aggregation, the resulting dataset contained 738 unique PD-L1 inhibitor structures represented by SMILES strings. The second dataset was obtained from a GitHub repository containing curated PD-L1 inhibitor compounds derived from patent data with reported pIC50 values. These two datasets were selected to enable comparison between a heterogeneous real-world bioactivity source and a cleaner, more standardized, curated dataset.

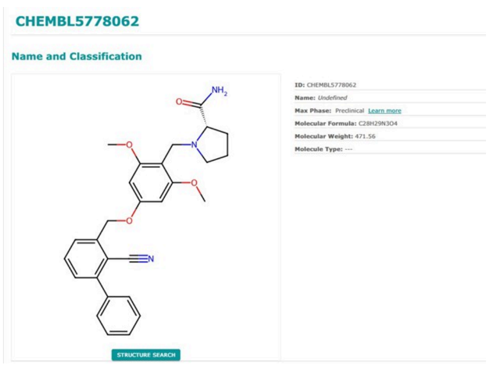

Fig. 1.Two-dimensional representation of CHEMBL5778062 with compound metadata.

The figure shows the molecular scaffold of CHEMBL5778062 and its associated compound information obtained from the ChEMBL database. The displayed structure provides a visual understanding of the compound's chemical framework, including ring systems and substituent groups, which are relevant for descriptor generation and activity prediction in machine learning-based QSAR modeling

1) *Data Preprocessing*

a) Converted IC50 values to pIC50

b) Removed duplicate molecules using SMILES

c) Aggregated repeated values using median

d) Standardized molecular structures

2) *Transforming IC50 to pIC50*

Our dataset shows that the IC50 distribution is skewed right, with most compounds exhibiting low nanomolar activity and a smaller subset shows high IC50 values. We decided to transform IC50 to pIC50 to stabilize variance and improve regression performance, which is exactly what the researchers in Tong et al followed.

IV. TABLE I : ORIGINAL DATA STATISTICS

TABLE I : PRESENTS THE STATISTICAL SUMMARY OF THE RAW IC50 VALUES.

Statistic	IC50(nM)
count	1770
mean	1428.047465
std	3117.893371
min	0.006
25%	11
50%	100
75%	555
max	20000

The large spread and high standard deviation indicate strong variability and skewness in the original activity data,

motivating the use of logarithmic transformation for improved modeling.

- As shown in Table I, the original IC50 values exhibited a very wide range and a high standard deviation, confirming that the raw activity data were highly skewed and variable.

TABLE II : : **SUMMARIZES THE LOG-TRANSFORMED IC50 VALUES EXPRESSED AS PIC50. THE TRANSFORMED DISTRIBUTION IS MORE NORMALIZED AND BETTER SCALED FOR REGRESSION ANALYSIS, MAKING IT MORE APPROPRIATE FOR PREDICTIVE MODELING.**

Column1	IC50_nM	pIC50
count	968	968
mean	2435.1901	6.357323
std	3853.40113	1.030798
min	0.006	4.989276
25%	300	5.967381
50%	505	6.296709
75%	1078	6.522879
max	10250	11.22185

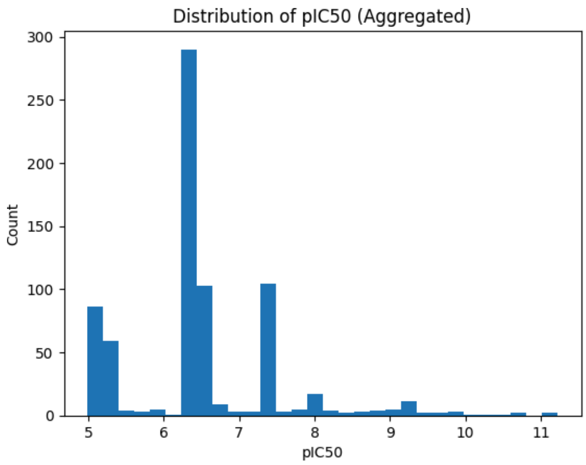

Fig. 2. Distribution of aggregated pIC50 values in the processed dataset.

This figure shows the frequency distribution of pIC50 values after preprocessing and aggregation. Most compounds are concentrated in the moderate activity range, while fewer samples appear at the extreme ends, indicating some remaining imbalance in the dataset.

After we log transformed IC50 values to pIC50, duplicate measurements for identical molecule structures were consolidated. Because multiple experimental IC50 values may exist for the same SMILES string across different documents or assays, the median plC50 value was computed for each unique molecule. This aggregation step reduces experimental noise and prevents data leakage arising from repeated structures appearing in both training and test sets. The resulting dataset consisted of 738 unique PD-L1 inhibitor structures.

1) *Descriptor Generation*

Molecular descriptors were generated using RDKit and Mordred to capture a broad range of chemical features from the PD-L1 inhibitor structures.. After converting each SMILES string into an RDKit molecular object, all 738 molecules were successfuly parsed, with no invalid SMILES identified. RDKit descriptors were first calculated, producing 217 descriptors for each molecule. In parallel, Mordred descriptors were generated. This step produced 1,613 Mordred descriptors per molecule. The RDKit and Mordred descriptor set were then combined, resulting in ad initial descriptor matrix of 738 x 1,830Initially, over 200 descriptors were extracted and filtered based on variance, correlation, and stability.

Several cleaning and filtering steps were applied to improve the quality of the descriptor matrix. First, infinite values and extremely large values were removed and replaced with NaN, and missing values were filled using the medican of each descriptor. Next, descriptors with more than 80% zero values were removed, reducing the feature space to 1,483 descriptors. A variance filter (variance <0.1) was then applied to remove low information feature, further reducing the dataset to 834 descriptors. Finally, highly correlated descriptors were removed using a Pearson correlation threshold of 0.90, resulting in a final feature matrix of 738 x 161. This step was important to reduce redundancy and improve model stability.After we log transformed IC50 values to pIC50, duplicate measurements for identical molecule structures were consolidated. Because

multiple experimental IC50 values may exist for the same SMILES string across different documents or assays, the median plC50 value was computed for each unique molecule. This aggregation step reduces experimental noise and prevents data leakage arising from repeated structures appearing in both training and test sets. The resulting dataset consisted of 738 unique PD-L1 inhibitor structures.

Molecular descriptors are quantitative numerical representations derived from chemical structures that encode physiochemical properties, topological characteristics, and other structure features relevant to biological activity. In our case, combining RDKit and Mordred allowed us to capture both general and more detailed structural information. The final cleaned dataset was organized so that each row represents a molecule and each column represents a descriptor, with pIC50 as the target variable. This dataset was then used for model training these descriptors quantitatively encode properties such as molecular weight, lipophilicity, polar surface area, hydrogen bond capacity, and electronic characteristics. Initially, we had 217 molecules descriptors. To ensure stability, we removed the extreme large value descriptor. After cleaning, 216 stable molecules were available for modeling.

$$R^2 = 1 - \frac{\sum\left(y_{true} - y_{pred}\right)^2}{\sum\left(y_{true} - \bar{y}\right)^2}$$

Fig. 1. Description

1) *Fingerprint Selection Rationale*

In the present study, descriptor-based modeling was prioritized because descriptors provide interpretable physicochemical, topological, and electronic features that can be directly linked to biological activity. This is particularly helpful in a QSAR setting where understanding

1) *Model Development*

Three machine learning regression models were used in this study: Random Forest, Gradient Boosting, and XGBoost. These methods were selected because of their ability to capture nonlinear relationships between molecular descriptors and biological activity. The dataset was divided using an 80:20 train-test split. The implementation was carried out in Python using Google Colab, with data preprocessing performed using pandas and NumPy, descriptor generation using RDKit and Mordred, model development using scikit-learn and XGBoost, and visualization using Matplotlib.

1) *Evaluation Metrics*

2) *Model performance was evaluated using the coefficient of determination (R2), root mean square error (RMSE), and mean absolute error (MAE). These metrics were chosen because they provide complementary views of regression performance. The R2 value measures the proportion of variance in the target variable explained by the model, while RMSE and MAE quantify prediction error, with lower values indicating better predictive accuracy. Together, these metrics enabled direct comparison of model performance across algorithms and datasets.*

3) *(a) R^2 (Coefficient of Determination)*
   a) Measures how well predicted values match actual pIC50 values.
   b) R^2 ranges from 0 to 1.

Higher R^2 indicates better predictive power.

1) $R^2 > 0.7$ → Strong predictive model
2) R^2 between 0.5–0.7 → Moderate model
3) $R^2 < 0.5$ → Weak model

1) *(b) RMSE (Root Mean Squared Error)*
   a) Measures the average prediction error.
   b) Penalizes large errors more heavily.
   c) Lower RMSE indicates better accuracy.

$$R^2 = 1 - \frac{\sum(y_{true} - y_{pred})^2}{\sum(y_{true} - \bar{y})^2}$$

Fig. 1. Description

$$R^2 = 1 - \frac{\sum(y_{true} - y_{pred})^2}{\sum(y_{true} - \bar{y})^2}$$

Fig. 1. Description

A. Train Random Forest Model

We trained a Random Forest(RF) model using 500 trees with a random state of 42 to ensure reproducibility. The model was fitted using 80% of the dataset as the training set, which included approximately 172 molecule descriptors. The remaining 20% of the data was used as the test set to evaluate the model's ability to predict molecules with strong affinity toward PD-L1. Model performance was assessed using standard regression metrics including root mean square error (RMSE), mean absolute error (MAE), and the coefficient of determination (((R^2).

The Random Forest model was evaluated by comparing the predicted pIC50 values with the experimentally observed values using a predicted-versus-actual plot, which provides a visual assessment of the model's predictive performance. In addition, feature importance analysis was performed to determine which molecular descriptors contributed most to the model's predictions. This analysis identified the top 15 descriptors that had the strongest influence on predicting PD-L1 binding affinity. These descriptors help highlight the key physicochemical properties associated with higher inhibitory activity and provide insight into the structural characteristics that may contribute to effective PD-L1 inhibition.

V. RESULTS AND DISCUSSION

Preliminary results indicate that predictive performance varied across both datasets and regression models. On the ChEMBL dataset, Random Forest achieved the strongest overall performance with

VI. RESULT 1

1) *Source Code*

This project was implemented in Python using Google Colab. Data preprocessing and manipulation are performed using pandas and NumPy. Molecular descriptors are generated from SMILES representations using RDKit followed by additional descriptor calculations using the Mordred package to expand the feature space and capture more comprehensive molecular properties. Three models were implemented using scikit learn to predict pIC50 values, and model performance is evaluated using R2, RMSE, and MAE. Visualization was performed using Matplotlib.X`

(c) MAE (Mean Absolute Error)

1) Measures the average absolute difference between predicted and true values.
2) Easier to interpret than RMSE.
3) Lower MAE is better.

, RMSE = 0.72, and MAE = 0.524, while Gradient Boosting and XGBoost showed slightly lower predictive accuracy. In contrast, the GitHub PD-L1 dataset produced better results for all tested models, with XGBoost achieving the highest performance at , RMSE = 0.356, and MAE dataset. These findings suggest that dataset consistency and curation quality have a substantial impact on QSAR performance and may influence predictive reliability as strongly as the choice of learning algorithm itself. Overall, the results support the use of curated bioactivity datasets for more stable and generalizable PD-L1 activity prediction.

To contextualize the present results, our best-performing GitHub model (XGBoost, $R^2 = 0.648$, RMSE = 0.356, MAE

= 0.257) was compared against the CatBoost benchmark reported by Tong et al., who achieved test-set performance

of $R^2 = 0.83$, RMSE = 0.22, and MAE = 0.17. This gap suggests that although the current framework shows promising predictive ability, further improvement is needed through stronger feature engineering, additional model optimization, and possibly the inclusion of fingerprint-based representations and explainability-guided descriptor selection.

1) *Model Performance on ChEMBL Dataset*

VII. TABLE II : DESCRIPTION OF TABLE

TABLE I : Description of Table

Model	RMSE	MAE	R^2
Random Forest	0.72	0.524	0.476
Gradient Boosting	0.732	0.512	0.459
XGBoost	0.737	0.531	0.452

The models showed moderate performance due to high variability and noise in the dataset.

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1) *Model Performance on GitHub Dataset*

VIII. TABLE III : DESCRIPTION OF TABLE

TABLE II : Description of Table

Model	RMSE	MAE	R^2
Random Forest	0.36	0.265	0.639
Gradient Boosting	0.368	0.270	0.625
XGBoost	0.356	0.257	0.648

The performance improved significantly due to cleaner and more consistent data.

1) *Comparative Analysis*

a) R^2 improved from ~0.45 → ~0.65

b) RMSE reduced from ~0.73 → ~0.36

c) XGBoost performed best across both datasets

While analyzing the results, we observed that the same models performed very differently across datasets. This highlighted how critical dataset quality is compared to just model selection.

1) *Feature Importance*

Key descriptors influencing activity

1) Hydrophobicity (MolLogP)

2) Surface Area (VSA descriptors)

3) Electronic properties (BCUT descriptors)

4) As illustrated in Fig. 5, a small subset of variables contributed disproportionately to prediction, indicating that model performance was driven mainly by a limited number of highly informative features.

1) In addition to R^2, our RMSE and MAE values remained higher than what would be expected from a highly optimized benchmark model, suggesting that further refinement is needed. This comparison shows that model architecture alone is not sufficient; dataset quality and descriptor relevance also strongly affect prediction accuracy.

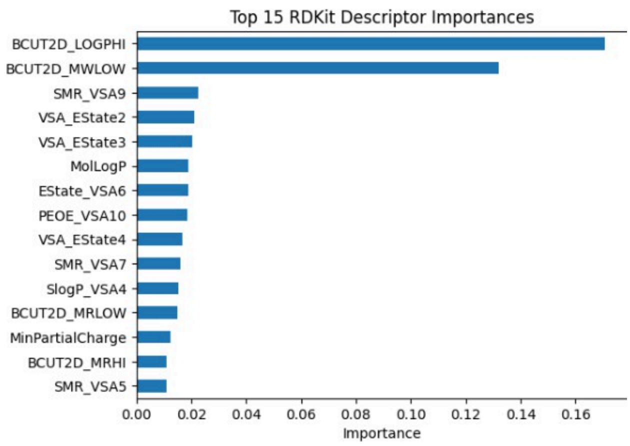


Fig. 4. Ranked importance scores of the top 15 RDKit descriptors in the developed QSAR framework.

The plot illustrates the relative contribution of key molecular descriptors to the predictive model. The dominance of BCUT2D_LOGPHI and BCUT2D_MWLOW suggests that hydrophobicity- and molecular size-related characteristics are critical for explaining variation in PD-L1 inhibitory potency, while the remaining descriptors provide supporting information related to surface area, electrotopological state, and partial charge distribution.

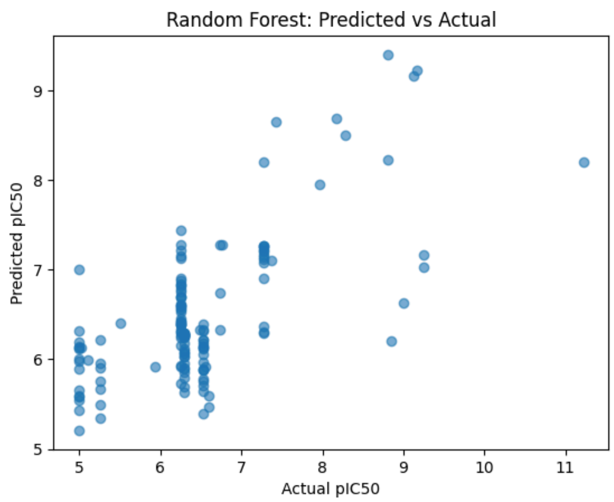


Fig. 5. Scatter plot of observed and predicted pIC50 values for the Random Forest-based QSAR model.

The plot provides a visual assessment of regression performance by comparing the experimental pIC50 values with model predictions. Although the Random

Forest model captures the overall trend in the data, the spread of points away from the ideal agreement pattern indicates residual prediction error, particularly for compounds in the higher activity range.

1) *Results of the Three Models and Comparison Between Datasets*

Three machine learning regression models, namely Random Forest (RF), Gradient Boosting Machine (GBM), and XGBoost (XGB), were evaluated on two PD-L1 inhibitor datasets: the ChEMBL dataset and the GitHub PD-L1 dataset. The results show that model performance varied considerably across the two datasets, indicating that dataset characteristics strongly influenced predictive accuracy.

For the ChEMBL dataset, the Random Forest model achieved the best performance among the three models with $R^2 = 0.476$, RMSE = 0.72, and MAE = 0.524.

The Gradient Boosting model followed closely with $R^2 = 0.459$, RMSE = 0.732, and MAE = 0.512.

The XGBoost model showed the lowest performance on this dataset with $R^2 = 0.452$, RMSE = 0.737, and MAE = 0.531. These values suggest that all three models had only moderate predictive capability on ChEMBL, likely because this dataset is more heterogeneous and contains higher experimental variability.

For the GitHub PD-L1 dataset, all three models performed better than they did on ChEMBL. In this case, XGBoostachieved the best overall results with $R^2 = 0.648$, RMSE = 0.356, and MAE = 0.257. The Random

Forest model performed nearly as well, with $R^2 = 0.639$, RMSE = 0.36, and MAE = 0.265. The Gradient Boosting model again ranked third, with $R^2 = 0.625$, RMSE

= 0.368, and MAE = 0.270. The improved results on this dataset indicate that the models benefited from the cleaner and more consistent structure of the GitHub data.

A direct comparison between the two datasets shows a clear improvement in performance on the GitHub dataset across all three models. For example, the best R^2 increased from 0.476 on ChEMBL to 0.648 on GitHub, while the RMSEdropped from about 0.72–0.74 on ChEMBL to about 0.36–0.37 on GitHub. Similarly, MAE values decreased from about 0.51–0.53 on ChEMBL to about 0.26–

0.27 on GitHub. This suggests that the GitHub dataset is easier for machine learning models to learn from, possibly because it contains less noise, a narrower activity range, and more structurally similar compounds. By contrast, the ChEMBL dataset represents a more realistic but more difficult prediction problem due to its higher diversity and variability.

Overall, the results indicate that XGBoost was the best-performing model across datasets, especially on the GitHub dataset, while Random Forest was slightly more robust on the ChEMBL dataset. Gradient Boosting showed competitive but slightly lower performance in both cases. These findings suggest that boosted ensemble methods are highly effective for curated QSAR datasets, whereas Random Forest may remain more stable for noisier experimental datasets.

```
results = []

for name, model in models.items():
    model.fit(X_train, y_train)
    pred = model.predict(X_test)

    rmse = np.sqrt(mean_squared_error(y_test, pred))
    mae = mean_absolute_error(y_test, pred)
    r2 = r2_score(y_test, pred)

    results.append({
        "Model": name,
        "RMSE": rmse,
        "MAE": mae,
        "R2": r2
    })

results_df = pd.DataFrame(results).sort_values("R2", ascending=False)
print(results_df)
```

```
                     Model      RMSE       MAE        R2
1  Gradient Boosting Machine  0.342765  0.251396  0.673662
0              Random Forest  0.346204  0.252276  0.667080
2                    XGBoost  0.350404  0.250266  0.658953
```

Fig.6. Preprocessing and fitting procedure of the XGBRegressor model. *The figure demonstrates the removal of object-type attributes from the input dataset prior to model fitting. After preprocessing, the XGBRegressor was trained on the numerical descriptor set, and the model configuration output verifies the applied hyperparameter settings used for regression analysis.*

1) As shown in Fig. 6, object-type columns were removed before training the XGBoost regressor so that the model operated only on numerical features.

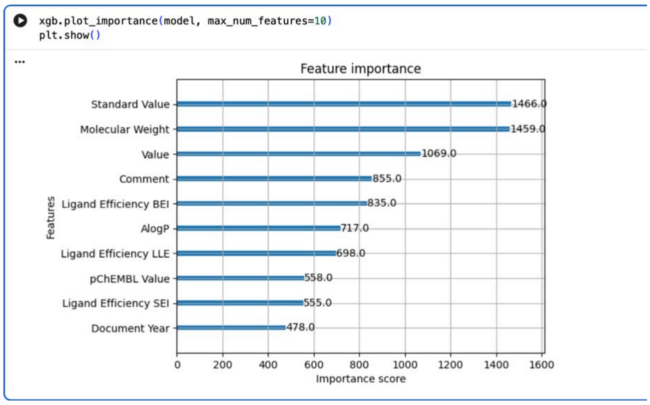
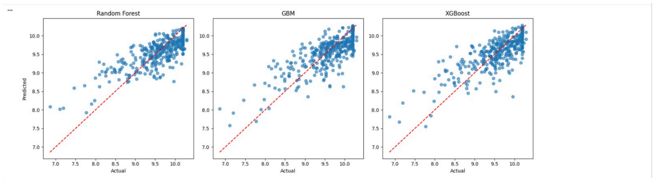


Fig. 7. Ranked feature importance scores obtained from the XGBoost regression model.

The plot illustrates the relative contribution of the top ten features used in the prediction task. The results indicate that a subset of descriptors and activity-related variables had a dominant influence on model performance, thereby improving the interpretability of the QSAR framework.

Fig. 8 summarizes the comparative regression performance of the evaluated ensemble models and shows that the overall differences between the methods were relatively small, although the best model varied depending on the dataset.




```
object_cols = X_train.select_dtypes(include='object').columns

X_train_processed = X_train.drop(columns=object_cols)
X_test_processed = X_test.drop(columns=object_cols)

model.fit(X_train_processed, y_train)
```

```
XGBRegressor
XGBRegressor(base_score=None, booster=None, callbacks=None,
             colsample_bylevel=None, colsample_bynode=None,
             colsample_bytree=0.8, device=None, early_stopping_rounds=None,
             enable_categorical=False, eval_metric=None, feature_types=None,
             feature_weights=None, gamma=None, grow_policy=None,
             importance_type=None, interaction_constraints=None,
             learning_rate=0.05, max_bin=None, max_cat_threshold=None,
             max_cat_to_onehot=None, max_delta_step=None, max_depth=6,
             max_leaves=None, min_child_weight=None, missing=nan,
             monotone_constraints=None, multi_strategy=None, n_estimators=500,
             n_jobs=None, num_parallel_tree=None, ...)
```

Fig. 9. *Actual versus predicted pIC50 values obtained from the Random Forest, GBM, and XGBoost regression models. The scatter plots illustrate the agreement between experimental and predicted values for each model, while the red dashed diagonal line represents the ideal 1:1 relationship. The results indicate that all three models achieved strong predictive performance, with Random Forest showing the closest clustering around the reference line and therefore the most consistent prediction accuracy within the QSAR framework.*

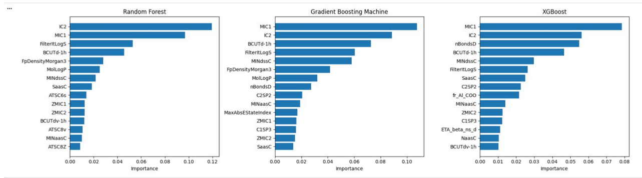


Fig. 10. *Ranked feature importance scores obtained from the Random Forest, Gradient Boosting Machine, and XGBoost regression models. The bar plots illustrate the relative contribution of the most influential molecular descriptors used in the prediction of pIC50 values. The results indicate that a small group of descriptors, including MIC1, IC2, and BCUT-based features, contributed most strongly to model performance across all three algorithms, thereby enhancing the interpretability and robustness of the QSAR framework.*

Our best-performing model on the curated GitHub dataset

achieved an R^2 of 0.648, whereas Tong et al. reported a CatBoost-based QSAR model with an R^2 of approximately

0.83. This indicates that although our ensemble models produced moderate-to-good predictive performance, there is still a clear gap compared with the benchmark study. The

difference may be due to variations in dataset curation, descriptor selection, feature engineering, and model optimization.

IX. CONCLUSION AND FUTURE WORK

This study presented a machine learning-based QSAR framework for predicting PD-L1 inhibitory potency using molecular descriptors derived from ChEMBL and GitHub datasets. The results showed that predictive performance depended strongly on dataset characteristics. Although all three models demonstrated moderate-to-good performance, the curated GitHub dataset consistently produced better results than the more heterogeneous ChEMBL dataset. Among the evaluated methods, XGBoost achieved the strongest overall performance, particularly on the GitHub dataset, whereas Random Forest showed relatively greater robustness on the noisier ChEMBL dataset.

These findings highlight the importance of dataset quality, descriptor relevance, and model selection in QSAR-based drug discovery. At the same time, the gap between the best model in this study and the higher benchmark performance reported by Tong et al. suggests that further improvement is still possible through more refined feature engineering, model optimization, and data curation. Tong et al. reported a CatBoost-based model with a test-set R^2 of 0.83, which provides a useful benchmark for future refinement of the present framework.

In future work, fingerprint-based representations such as Morgan and MACCS keys will be evaluated in addition to descriptor-based features. Morgan fingerprints are suitable for PD-L1 inhibitors because they capture circular substructures and local chemical environments, which are important for molecules containing diverse aromatic scaffolds, heterocycles, and substituted ring systems. By contrast, MACCS keys provide a smaller and more interpretable predefined structural vocabulary, which may be useful for identifying broad recurring motifs but may capture less structural detail. Therefore, comparing Morgan and MACCS fingerprints will help determine whether richer substructure encoding improves predictive performance for the chemically diverse PD-L1 inhibitor space.

X. AUTHOR CONTRIBUTIONS

Binh Diep was responsible for dataset preprocessing, data cleaning, and initial model implementation. Sai Charitha Gopa contributed to descriptor generation, model training, evaluation, comparative analysis, and manuscript drafting. Rishik Kondura contributed to the literature review, interpretation of results, document formatting, and final editing. All authors reviewed and approved the final manuscript.

XI. .REFERENCES

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